Hein Htet

Embodied AI & Robotics Engineer

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Professional Summary

Robotics and embodied-AI engineer working across learned models and physical hardware. AI architecture lead and team co-lead at Magick Mind, responsible for the agent platform and the runtime that gives robots and wearable devices an on-platform mind. Previously project lead for a **12-DOF** bipedal humanoid, taking a reinforcement-learning locomotion policy from simulation onto hardware. Research background in self-supervised representation learning and sequence modeling over learned latents.

Education

King Mongkut's University of Technology Thonburi July 2025
B.Eng. Electronic and Infocommunication Engineering Bangkok, Thailand

- GPA **3.71/4.00**, First Class Honours. Coursework in machine learning, deep learning, and control.

Skills

- **Robotics & Embodied AI:** observation and action space design, Isaac Sim, MuJoCo, Zenoh, sim-to-real transfer, visual SLAM, ROS 2, real-time control
- **Machine Learning:** PyTorch, TorchScript, ONNX Runtime, self-supervised and generative modeling, reinforcement learning, model benchmarking
- **Agentic Systems:** agent runtime and harness design, skill systems with trust gating, tool use and delegation, multi-agent coordination, agent-assisted development, real-time voice agents, long-term memory
- **Programming:** Python, C++, C, Go, Rust · agentic coding with Claude Code and OpenCode
- **Infrastructure:** Hatchet, Kubernetes, Docker, gRPC, Centrifugo, CI/CD, Git
- **Hardware & Interfaces:** Jetson Orin Nano, ESP32, STM32, CAN (SocketCAN), IMU, LiDAR
- **Language:** English — TOEIC **970/990**

Work Experience

Magick Mind April 2026 – Present
AI Architecture Lead & Team Co-Lead Bangkok, Thailand

- Lead AI architecture and co-lead the team across a **17-service** platform spanning companion agents, memory, voice, and robotics.
- Built the real-time voice runtime that gives ESP32-class robots and wearables an on-platform mind — streaming speech recognition, persona-driven reasoning, and synthesis with barge-in, at **50 concurrent sessions**.
- Architected Mindroid, an open-source Rust agent SDK with a gated skill system and feature-gated builds, so the same runtime runs in the cloud or on constrained edge hardware.
- Designed the pairing and token-broker architecture that binds a device to its own persona-backed agent identity, keeping tenant credentials off the hardware.
- Designed a deterministic recursive language-model architecture that replaces open-ended reasoning loops with a fixed combinator chain, validated against reference implementations across **four** public reasoning benchmarks.
- Building a Zenoh-based robotics adapter that exposes robot capabilities as agent tools, so platform agents can execute missions on a quadruped companion robot and a robotic manipulator; in development, validated in simulation.

Magick Robotics March 2025 – March 2026
Project Lead & Senior Robotics Engineer Bangkok, Thailand

- Designed the observation and action spaces for the Tet Hlan **12-DOF** bipedal humanoid, then took its reinforcement-learning locomotion policy from simulation onto hardware at **50 Hz**.
- Led software architecture across embedded systems, reinforcement-learning locomotion, and mechanical integration.
- Built the real-time sensing and motor control stack beneath the policy, and migrated the platform to ROS 2.

KMUTT Deep Learning Lab August 2022 – June 2025
Research Assistant Bangkok, Thailand

- Designed a self-supervised generative model for speech conversion in PyTorch — learned latent representations with a neural vocoder — outperforming **two** published baselines.
- Trained and benchmarked sequence models over continuous learned latent representations, and designed the evaluation protocol used to compare them.

Panasonic Research Lab October 2023 – July 2024
Student Researcher Singapore

- Rewrote ORB-SLAM2 for constrained robot compute, implementing its numerical and bundle-adjustment libraries from scratch, and ported a visual localization pipeline to dependency-free C.

Mu Space Corp February 2022 – July 2022
Embedded Engineering Intern Bangkok, Thailand

- Developed satellite firmware across **five** STM32 architectures, including an ion thruster control subsystem.

Honors & Leadership

- Gold Prize, Student Innovation Challenge Thailand 2024 · Founder, Clash of Robots, Myanmar's first national robotics competition · Mindroid (Rust), open-source author